



Birla Institute of Technology & Science, Pilani
Work-Integrated Learning Programmes Division
MTech in Artificial Intelligence and Machine Learning
S2_2025-2026, AIMLCZG557/AECLZG557

Assignment 1 – PS10 - Grid Travel Agent - [Weightage 12%]

Read through this entire document very carefully before you start!

Problem statement

Assume a queen in distress on a grid at a specified location. The knight is at another location. Help the knight agent to reach the queen for help. The problem is leading the knight agent towards the queen through the grid that has obstacles like water bodies, forest, fire, wild animals and mountain terrain. The knight has to take the shortest path through the grid. He earns 5 rewards each time he passes near water bodies and forest. 5 rewards are to be taken away from the knight while he passes near fire, and wild animals and 3 awards to be taken away if he passes near mountain terrain.

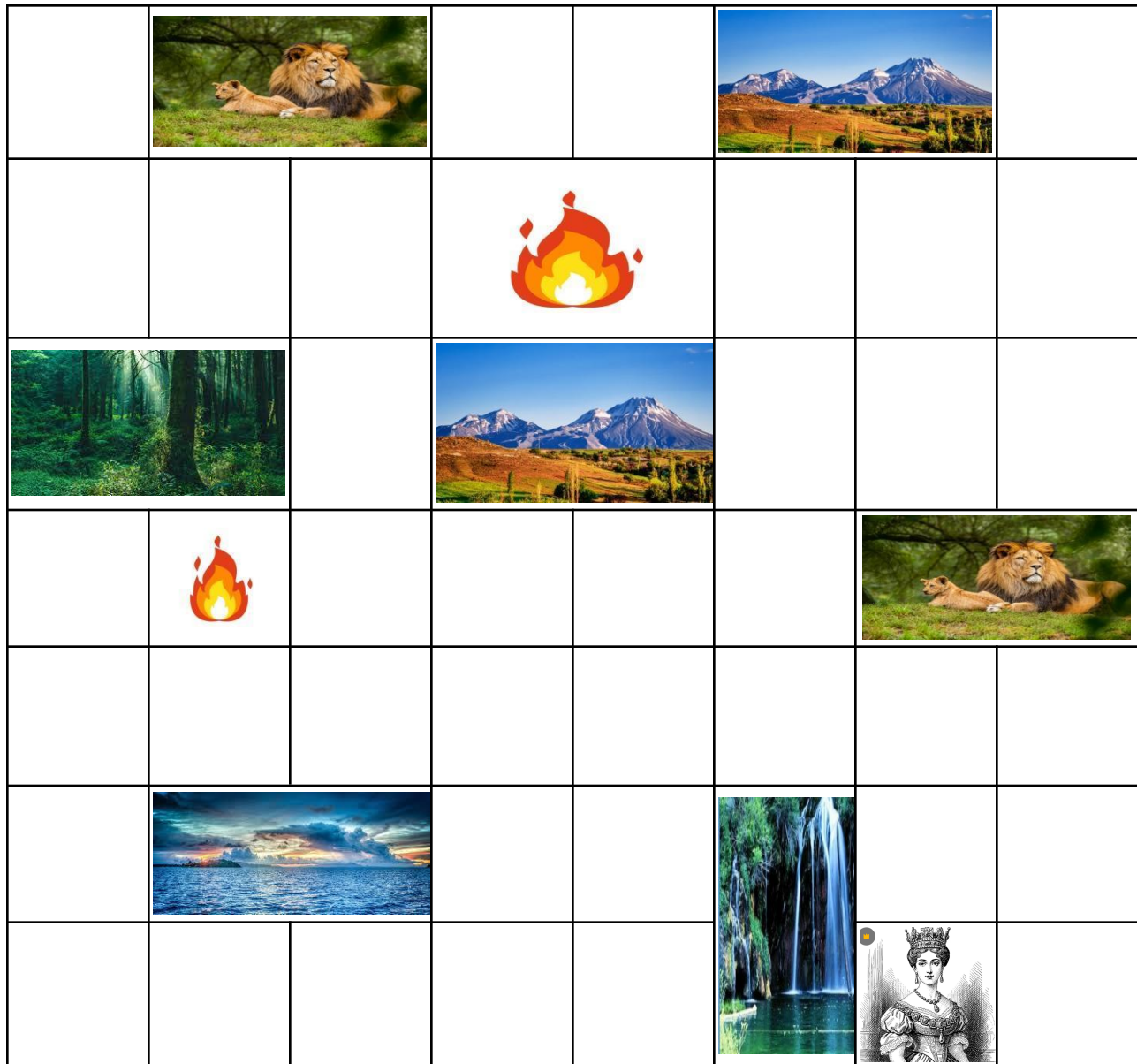
The following figure gives the initial grid positions. The desired solution must have maximum awards but must have travelled through a smaller number of squares. The environment is fully observable.

Use the A* algorithm to solve the problem.

Note :

Initial Board configuration





1. Explain the PEAS (Performance Measure, Environment, Actuators, Sensors) for the routing agent.
2. Explain the environmental characteristics for the given problem.
3. Explain the heuristic function used for the problem
4. Use appropriate data structures and implement the A* Search algorithm.
5. The starting point is to be obtained from the user as input.
6. Print the space and time complexity as part of your implementation. Do not calculate theoretically.

Sample input and output

Sample Input:

The grid using appropriate data structure

Starting point is fixed (0,0)

Sample Output:

The path taken by the knight and the cost (rewards)

Eg. (0,0), (1,0), (1,1), (1,2), (2,2), (3,2), (5,2),(5,3), (5,4),(5,5), (5,6), (6,6)(7,6)

The cost of the path is: 10

Deliverables:

- PDF document **designPSXX <group id>.pdf** detailing your solution.
- **[Group id] _Contribution.xlsx** mentioning the contribution of each student in terms of percentage of work done. Columns must be “Student Registration Number”, “Name”, “Percentage of contribution out of 100%”. If a student did not contribute at all, it will be 0%, if all contributed then 100% for all.
- **inputPSXX.txt** file used for testing
- **outputPSXX.txt** file generated while testing
- .py file containing the python code. Create a single notebook. Do not fragment your code into multiple files
- Zip all of the above files including the design document in a folder with the name:
 - **[Group id] _A1_PSXX_XXXXXXXXXX.zip** and submit the zipped file.
 - Group Id should be given as **Gxxx** where **xxx** is your group number. For example, if your group is 26, then you will enter **G026** as your group id.

Instructions:

- It is compulsory to make use of the data structure(s) / algorithms mentioned in the problem statement.
- Ensure that all data structure insert and delete operations throw appropriate messages when their capacity is empty or full. Also ensure basic error handling is implemented.
- For the purposes of testing, you may implement some functions to print the data structures or other test data. But all such functions must be commented before submission.
- Make sure that you read, understand, and follow all the instructions

- Ensure that the input, prompt and output file guidelines are adhered to. Deviations from the mentioned formats will not be entertained.
- The input, prompt and output samples shown here are only a representation of the syntax to be used. Actual files used to evaluate the submissions will be different. Hence, do not hard code any values into the code.
- Please note that the design document must include:
 - One alternate way of modeling the problem with the performance implications.
 - Writing a good technical report and well documented code is an art. Your report cannot exceed 4 pages. Your code must be modular and quite well documented.
 - You may ask queries in [this dedicated discussion section](#). Beware that only hints will be provided and queries asked **in other channels like MS Teams, emails, Taxila inbox will not be responded to.**

Deadline

1. The strict deadline for submission of the assignment is **8th June 2026 11:55 PM IST (Monday)**.
2. The deadline has been set considering extra days from the regular duration in order to accommodate any challenges you might face. No further extensions will be entertained.
3. Late submissions will be evaluated as below. Submissions made between:
 - a. 8th June 2026 11:56PM to 9th June 2026 11:55PM → Deduct 2M for late submission and evaluation for 10 marks.
 - b. 9th June 2026 11:56PM to 10th June 2026 11:55PM → Deduct 6M for late submission and evaluation for 6marks.
 - c. Beyond 10th June 2026 11:55PM → No evaluations, 0 marks will be awarded.

How to submit

- This is a group assignment.
- Each group has to make one submission (only one, no resubmission) of solutions.
- Each group should zip all the deliverables in one zip file and name the zipped file as [Group id]_A1_PSXX_XXXXXXXXXX.zip and submit the zipped file.
- Group Id should be given as Gxxx where xxx is your group number. For example, if your group is 26, then you will enter G026 as your group id.
- Assignments should be submitted via Taxila > Assignment section. Assignments submitted via other means like email etc. will not be graded.

Evaluation

- The assignment carries **12 Marks**.
- Grading will depend on:
- Fully executable code with all functionality working as expected
 - Well-structured and commented code
 - Accuracy of the design document.
 - Every bug in the functionality will have negative marking.
 - Marks will be deducted if your program fails to read the input file used for evaluation due to change / deviation from the required syntax.
- We encourage students to take the upcoming assignments and examinations seriously and submit only original work. Please note that plagiarism in assignments will be taken seriously. Refer to the detailed policy [here](#).
- Source code files which contain compilation errors will get at most 25% of the value of that question.

Readings

Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell, 4th Edition.